SULIT

UTM UNIVERSITI TEKNOLOGI MALAYSIA | Fakulti Komputeran

UNIVERSITI TEKNOLOGI MALAYSIA
MIDTERM EXAMINATION SEMESTER II 2024 / 2025

SUBJECT CODE	: SECR2043
SUBJECT NAME	: OPERATING SYSTEMS
SECTION	: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, MJIIT, SPACE
TIME	: 08:00 PM to 10:30 PM
DATE/DAY	: 22 MAY 2025 (Thursday)
VENUES	: L50, N24, MJIIT

INSTRUCTIONS:

Answer **ALL objective questions** in the **Answer Sheet** provided on **page 7**.
Answer **ALL structured questions** in the **Appendix** provided on **pages 14 to 22**.
This exam will contributes **20%** towards the total marks of 100%.

(Please Write Your Lecturer Name and Section)

Name	
Matric No.	
Year / Course	
Section	
Lecturer Name	

This question booklet consists of 22 pages including the front page.

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PART A: OBJECTIVES [25 MARKS]

Answer ALL objective questions in the Answer Sheet provided on page 7.

1. In a non-preemptive scheduling environment, the CPU can be allocated to another process:

   A. Whenever a higher priority process arrives.
   B. Only after the current process terminates or enters the waiting state.
   C. At the expiration of every time quantum.
   D. At the discretion of the dispatcher.

2. Which of the following factors contributes to dispatcher latency?

   A. Saving the context of the current process and loading the context of the next process.
   B. Executing user-level instructions.
   C. Waiting for I/O operations to complete.
   D. Fetching program instructions from main memory.

3. High throughput in a computing system refers to:

   A. Individual processes completing in minimal time.
   B. A larger number of processes completing per unit time.
   C. Minimizing process waiting times.
   D. Achieving constant 100% CPU utilization.

4. *In a video rendering studio, large rendering jobs are submitted by various designers. To ensure each designer gets their finished videos back quickly, which scheduling goal should be emphasized?*

   A. Minimize waiting time
   B. Minimize turnaround time
   C. Maximize CPU utilization
   D. Minimize response time

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n of the following is true about First Come First Serve (FCFS) scheduling?

A. It favours short jobs.

B. It can cause the convoy effect.

C. It gives priority to the process with the highest priority.

D. It minimizes average waiting time.

6. Which algorithm gives each process a fixed time slot to run in cyclic order?

A. First Come First Serve

B. Shortest Job First

C. Round Robin

D. Priority Scheduling

7. Which of the following can help prevent starvation in priority scheduling?

A. Preemption

B. Round Robin

C. Aging

D. Shortest Remaining Time First

8. When implementing a multilevel queue with fixed priority scheduling, which queue runs first?

A. Highest priority queue

B. Lowest priority queue

C. Randomly selected queue

D. Queue with least waiting time

9. What happens when multiple processes update a shared variable without synchronization?

A. Deadlock

B. Consistent data

C. Race condition

D. Paging fault

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10. In which scenario is synchronization not necessary?

A. Two processes using a shared variable

B. A process accessing its own memory

C. Multiple threads updating a shared counter

D. Reading and writing to a file by two processes

11. Bounded waiting prevents:

A. Simultaneous execution

B. Starvation

C. Deadlock

D. Paging

12. Which of the following is NOT a requirement for the critical-section problem?

A. Fair scheduling

B. Mutual exclusion

C. Progress

D. Bounded waiting

13. Which of the following is true about Peterson's solution?

A. It uses semaphores

B. It's a software-based synchronization method

C. It requires kernel-level privileges

D. It uses spinlocks

14. Which of the following operations in Peterson's algorithm must be performed atomically to ensure correctness?

A. flag[i] = true

B. turn = j

C. Both A and B

D. None

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15. Which factor most limits the use of a test-and-set lock on every hardware platform?

A. It needs an atomic test-and-set CPU instruction that some processors do not provide.

B. It requires disabling interrupts each time the lock is checked.

C. It works only under co-operative (non-preemptive) multitasking.

D. It enforces strict FIFO order, which certain real-time systems reject.

16. Which critical section requirement is not guaranteed by Test-and-Set?

A. Mutual exclusion

B. Bounded waiting

C. Atomic access

D. Lock acquisition

17. Select true statements based on **Table 1** for the values of the control variables in the Peterson algorithm.

Table 1: Values for the Peterson's Algorithm Variables

Flag[0]	Flag[1]	turn
False	True	0

I. Process 0 gives way for Process 1 to enter the critical section.

II. Process 1 wants to enter the critical section.

III. Process 1 is allowed to enter the critical section.

IV. Process 0 is allowed to enter the critical section.

A. I and II

B. II and III

C. III and IV

D. I and IV

18. Mutex Locks is a special case of a Semaphore when _____ .

A. The Semaphore takes integer between 0 and 10.

B. The Semaphore integer is larger than 0.

C. The Semaphore integer is unrestricted.

D. The Semaphore integer is either 0 and 1.

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19. **Figure 1** shows the test and set function. Which of the following are true in terms of the output of the function.

```
boolean test_and_set (boolean *target) {
    boolean rv = *target;
    *target = TRUE;
    return rv:
}
```

Figure 1: Test and Set Function

I. The function only flips the value of *target if it is initially False

II. The function always flips the value of *target

III. The function return the modified value of *target

IV. The function returns the original value of *target

A. I and II

B. II and III

C. III and IV

D. I and IV

20. Select the correct sequence of execution for statements S1 and S2 based on the conditions described in **Figure 2**.

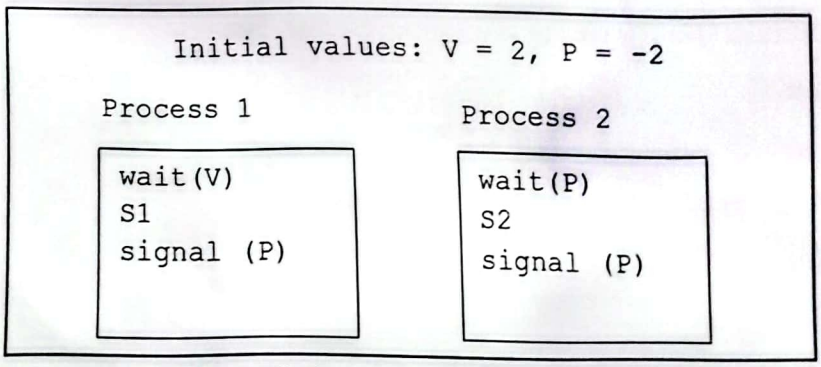

Figure 2: Semaphore of Process 1 and 2

A. S1 -> S2 -> S1 -> S2

B. S1 -> S2 -> S2 -> S2

C. S1 -> S1 -> S1 -> S2

D. S1 -> S1 -> S2 -> S2

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Section	
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Answer sheet for objective questions:

1		**11**	
2		**12**	
3		**13**	
4		**14**	
5		**15**	
6		**16**	
7		**17**	
8		**18**	
9		**19**	
10		**20**	

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PART B: STRUCTURED QUESTIONS [75 MARKS]

Answer ALL structured questions in the provided Appendix on Page 14 to 22.

QUESTION 1 [15 MARKS]

* Your answers for **Question 1** must be written in ***Appendix A***.

Table 2 below shows the trace addresses for three processes: P, Q, and R, as well as the address of the dispatcher.

Table 2: Trace addresses for Process P, Q, R and the dispatcher.

Process	Address
P	7000-7019
Q	8000-8012
R	10000-10019
Dispatcher	400-404

At instruction cycle 5, process P requests an I/O operation that will take 25 instruction cycles to complete. At cycle 35, process Q similarly requests an I/O operation lasting 15 instruction cycles. All processes, including the dispatcher, are allowed to execute for a maximum of 6 instruction cycles each time they are scheduled.

a) Draw a Gantt chart to trace the execution of all processes, including the dispatcher, for 90 instruction cycles. Clearly show the states of the processes (running, ready, or blocked) and the dispatcher throughout the timeline. **[12 marks]**

b) Provide the state of each process at instruction cycle 85, along with the number of instruction cycles remaining for each process. **[3 marks]**

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QUESTION 2 [15 MARKS]

* Your answers for **Question 2** must be written in ***Appendix B***.

A computer system is running five tasks: A (Print job), B (System update), C (Video rendering), D (Email synchronization), and E (Backup process). These tasks arrive at the times shown in **Table 3** below. Each task requires CPU Burst time and the system uses priority levels to manage them, where a lower number indicates a higher priority.

You are asked to evaluate how the system schedules these tasks using two scheduling strategies: First Come First Serve (FCFS) and Preemptive Priority Scheduling.

Table 3: Process arrival time, CPU Burst, and priority.

Process	Arrival Time (ms)	CPU Burst (ms)	Priority
A	0	6	3
B	1	8	1
C	2	7	4
D	3	3	2
E	5	5	5

a) Draw the Gantt chart showing the execution order of the processes using the First Come First Serve (FCFS) scheduling algorithm. [4 marks]

b) Draw the Gantt chart showing the execution order of the processes using the Preemptive Priority Scheduling algorithm. [4 marks]

c) Calculate the turnaround time and waiting time for each process using the Table 4 template below for FCFS and Preemptive Priority Scheduling. [5 marks]

Table 4: Turnaround Time and Waiting Time Template

Process	Arrival Time (ms)	CPU Burst (ms)	Completion Time (ms)	Turnaround Time (ms)	Waiting Time (ms)
A	0	6			
B	1	8			
C	2	7			
D	3	3			
E	5	5			

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d) Based on your calculations, which scheduling algorithm achieves better overall performance? Compare and justify your answer by referring to the average turnaround time and average waiting time for each algorithm. [2 marks]

QUESTION 3 [15 MARKS]

* Your answers for **Question 3** must be written in ***Appendix C.***

a) Consider the following Process P and Q. The initial values of shared variables are process[0] = FALSE, process[1] = FALSE, turn = 0 and n = 0

```
do{
    process[0] = TRUE;
    turn = 1;
    while(process[1] && turn == 1)
    {
        cout<<n;
    }
    process[0] = FALSE;
}while(TRUE);
```

Process P

```
do{
    process [1] = TRUE;
    turn = 0;
    while(process[0] && turn == 0)
    {
        n++;
        cout<<n;
    }
    process[1] = FALSE;
}while(TRUE);
```

Process Q

Figure 3: Peterson's Algorithm of Process P and Q

i. Please fill all the events output for (a), (b), (c) and (d) from the processes above in the given **Table 5** below. [8 marks]

Table 5: Peterson Algorithm Execution Trace

Time	P	Q	turn	Events
T_0	TRUE	FALSE	1	P request to enter critical section/ P enters critical section
T_1	TRUE	TRUE	0	*(a)*
T_2	FALSE	TRUE	0	P enters remainder section/ Q enters critical section
T_3	FALSE	TRUE	0	Q executes critical section
T_4	TRUE	TRUE	1	*(b)*
T_5	TRUE	TRUE	1	Q executes critical section/ P busy waiting in loop

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Time	P	Q	turn	Events
T_6	TRUE	FALSE	1	*(c)*
T_7	TRUE	TRUE	0	P executes critical section/ Q request to enter critical section
T_8	FALSE	TRUE	0	*(d)*
T_9	FALSE	TRUE	0	Q executes critical section
T_{10}	FALSE	TRUE	0	Q executes critical section

ii. What is the output at T_2 and T_9? **[2 marks]**

iii. What is the sequence of output produced by the processes if execution continues until T_{10}? **[2 marks]**

b) The Test-and-Set instruction is commonly used in synchronization mechanisms to solve the critical-section problem.

i. Briefly explain how the Test-and-Set instruction works. **[1 mark]**

ii. Write a short pseudocode example that uses Test-and-Set to implement mutual exclusion for two processes. **[2 marks]**

QUESTION 4 [15 MARKS]

*** Your answers for Question 4 must be written in *Appendix D***

Two processes, P_0 and P_1, cooperate to increment a shared variable S that is initially 0. They use the following **Mutex Lock pseudocode** to ensure mutual exclusion. Each process executes the following code fragment in **Figure 4 below.**

```
do {
    acquire(mutex);
    S = S + 3;
    release(mutex);
} while (condition);
```

Process P_0

```
do {
    acquire(mutex);
    S = S * 2;
    release(mutex);
} while (condition);
```

Process P_1

Figure 4: Mutex Lock of Process P_0 and P_1

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The **mutex** variable is initially **unlocked (0)**. The acquire (mutex) operation sets the mutex to 1 (locked) if it is not already locked; otherwise, the process waits. The release (mutex) operation sets the mutex back to 0 (unlocked).

a) Complete the following **Table 6** to show the operation and the value of mutex and S after each step. Fill in the blank cells *(a -l)*. **[12 marks]**

Table 6: Mutex Lock Execution Table

Step	Process	Operation	mutex	S
1	P_0	acquire(mutex)	1	0
2	P_0	*(a)*	1	*(b)*
3	P_1	acquire(mutex)	*(c)*	*(d)*
4	P_0	*(e)*	*(f)*	*(g)*
5	P_1	*(h)*	*1*	*(i)*
6	P_1	*(j)*	*(k)*	*(l)*

b) At which step is P1 **blocked** or **waiting** due to the mutex? **[1 mark]**

c) What is the final value of S after step 6? **[2 marks]**

QUESTION 5 [15 MARKS]

* Your answers for **Question 5** must be written in ***Appendix E***

Producer	Consumer

do {
wait(empty)
wait(mutex)
//add elements to the buffer
signal(mutex)
signal(full)
}
	do {
	wait(full)
	wait(mutex)
	//add elements to the buffer
	signal(mutex)
	signal(full)
	}

Figure 5: Semaphore Algorithm for Producer and Consumer

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a) Provide the initial values of **empty, full and mutex**, such that the Producer will not insert elements into the buffer when it is full and the Consumer will not retrieve elements from the buffer when it is empty. Assume the size of the buffer is 10 and the Producer-Consumer program just started. Refer to the Semaphore Algorithm for Producer and Consumer in **Figure 5** and fill in the appropriate initial values in **Table 7** below. [3 marks]

Table 7: Initial Values

Empty	Full	Mutex
(a)	*(b)*	*(c)*

b) Complete **Table 8** below based on the current values of empty, full and mutex. Assume that the sequence of execution is producer, producer and consumer. [12 marks]

Table 8: Semaphore Execution Table

Process	**Wait(empty)**	**Wait(mutex)**	**Signal(mutex)**	**Signal(full)**
Current value	5	1	-	5
Producer	*(a)*	*(b)*	*(c)*	*(d)*
Producer	*(e)*	*(f)*	*(g)*	*(h)*
Process	**Wait(full)**	**Wait(mutex)**	**Signal(mutex)**	**Signal(empty)**
Consumer	*(i)*	*(j)*	*(k)*	*(l)*

- End of Questions –

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APPENDIX A (ANSWER FOR QUESTION 1)

ANSWER FOR QUESTION 1 (b)

State of P = _____ Remaining Instruction Cycles for P = _____

State of Q = _____ Remaining Instruction Cycles for Q = _____

State of R = _____ Remaining Instruction Cycles for R = _____

ANSWER FOR QUESTION 2 (b)

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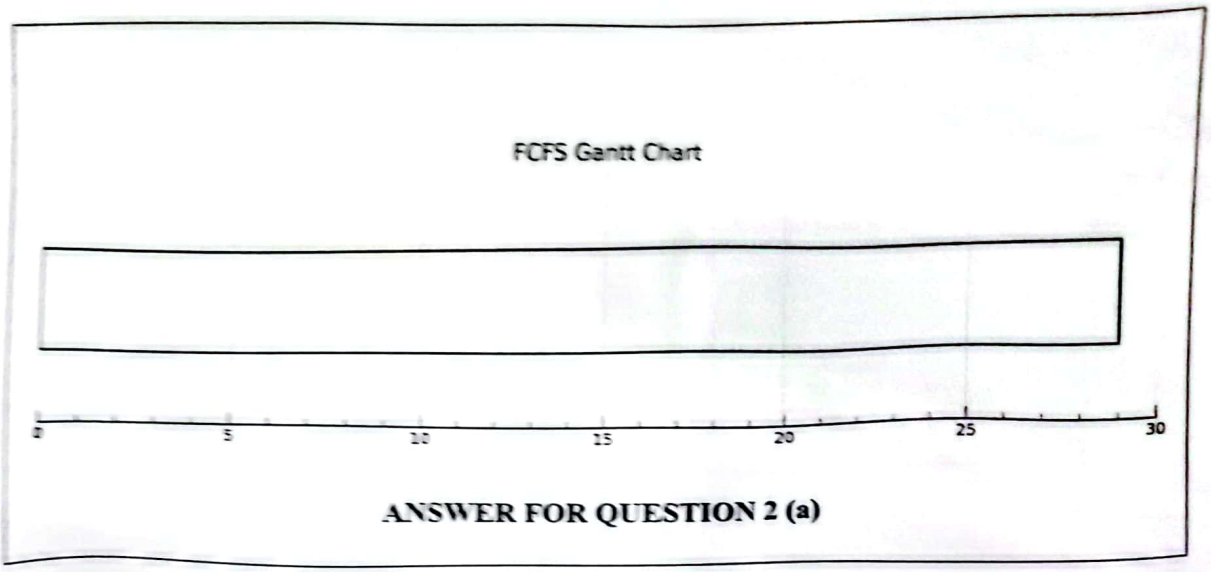
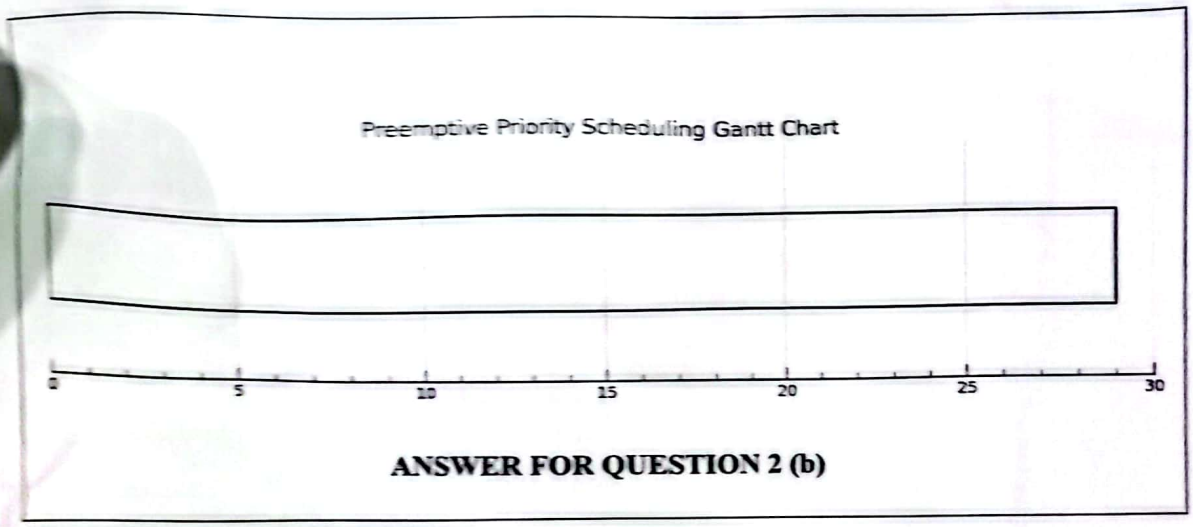
APPENDIX B (ANSWER FOR QUESTION 2)



ANSWER FOR QUESTION 2 (a)



ANSWER FOR QUESTION 2 (b)

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Table 4: Turnaround Time and Waiting Time (FCFS)

Process	Arrival Time (ms)	CPU Burst (ms)	Completion Time (ms)	Turnaround Time (ms)	Waiting Time (ms)
A	0	6			
B	1	8			
C	2	7			
D	3	3			
E	5	5			
			Average		

Table 4: Turnaround Time and Waiting Time (**Preemptive Priority Scheduling**)

Process	Arrival Time (ms)	CPU Burst (ms)	Completion Time (ms)	Turnaround Time (ms)	Waiting Time (ms)
A	0	6			
B	1	8			
C	2	7			
D	3	3			
E	5	5			
			Average		

ANSWER FOR QUESTION 2 (c)

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Average Turnaround Time FCFS = _____

Average Waiting Time FCFS = _____

Average Turnaround Time Preemptive Priority Scheduling = _____

Average Waiting Time Preemptive Priority Scheduling = _____

Based on the calculation above, scheduling algorithm that achieves better overall performance is _____

Justification:

_____

_____

_____

ANSWER FOR QUESTION 2 (d)

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APPENDIX C (ANSWER FOR QUESTION 3)

Table 5: Peterson Algorithm Execution Trace

Time	P	Q	turn	Events
T_0	TRUE	FALSE	1	P request to enter critical section/ P enters critical section
T_1	TRUE	TRUE	0	(a)
T_2	FALSE	TRUE	0	P enters remainder section/ Q enters critical section
T_3	FALSE	TRUE	0	Q executes critical section
T_4	TRUE	TRUE	1	(b)
T_5	TRUE	TRUE	1	Q executes critical section/ P busy waiting in loop
T_6	TRUE	FALSE	1	(c)
T_7	TRUE	TRUE	0	P executes critical section/ Q request to enter critical section
T_8	FALSE	TRUE	0	(d)
T_9	FALSE	TRUE	0	Q executes critical section
T_{10}	FALSE	TRUE	0	Q executes critical section

ANSWER FOR QUESTION 3 (a) i

Output at T_2 = _____

Output at T_9 = _____

ANSWER FOR QUESTION 3 (a) ii

Output sequence from T_0 to T_{10} = _____

ANSWER FOR QUESTION 3 (a) iii

Brief explanation of how the Test-and-Set instruction works:

_____

_____

_____

_____

_____

ANSWER FOR QUESTION 3 (b) i

Write a short pseudocode example that uses Test-and-Set to implement mutual exclusion for two processes:

ANSWER FOR QUESTION 3 (b) ii

APPENDIX D (ANSWER FOR QUESTION 4)

Table 6: Mutex Lock Execution Table

Step	Process	Operation	mutex	S
1	P_0	acquire(mutex)	1	0
2	P_0	*(a)* _____	1	*(b)* _____
3	P_1	acquire(mutex)	*(c)* _____	*(d)* _____
4	P_0	*(e)* _____	*(f)* _____	*(g)* _____
5	P_1	*(h)* _____	*1*	*(i)* _____
6	P_1	*(j)* _____	*(k)* _____	*(l)* _____

ANSWER FOR QUESTION 4 (a)

Step of P1 blocked or waiting due to the mutex = _____

ANSWER FOR QUESTION 4 (b)

Final value of S after step 6 = _____

ANSWER FOR QUESTION 4 (c)

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APPENDIX E (ANSWER FOR QUESTION 5)

Table 7: Initial Values

Empty	Full	Mutex
(a) _____	(b) _____	(c) _____

ANSWER FOR QUESTION 5 (a)

Table 8: Semaphore Execution Table

Process	Wait(empty)	Wait(mutex)	Signal(mutex)	Signal(full)
Current value	5	1	-	5
Producer	(a) _____	(b) _____	(c) _____	(d) _____
Producer	(e) _____	(f) _____	(g) _____	(h) _____
Process	**Wait(full)**	**Wait(mutex)**	**Signal(mutex)**	**Signal(empty)**
Consumer	(i) _____	(j) _____	(k) _____	(l) _____

ANSWER FOR QUESTION 5 (b)

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